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Faculty of Engineering, Built Environment and Information Technology

Fakulteit Ingenieurswese, Bou-omgewing en
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Tikologo ya Kago le Theknolotši ya Tshedimošo

Unit 8: Project Phases, Deliverables and Work Breakdown Structure

BSS 310 Engineering Management
2026

Lecturer: Prof. Michael Ayomoh



Quick Recap/Questions arising from Previous Lecture

The lecture addressed:

- Fundamentals of project scope and scope statement; the significance of a project scope; project stakeholders and the relationship between project scope and stakeholders; Stakeholders map and adaptation of all of these to the semester project.

Learning Outcomes for the Current Lecture

The expected learning outcomes at the end of this lecture include:

- Understanding the concepts of project, phases and deliverables.
- Understanding the principles and purpose of a Work Breakdown Structure (WBS) and being able to analyse, evaluate, and create one
- Understanding and being able to analyse, evaluate and create a Project Responsibility Assignment Matrix

Presentation Outline

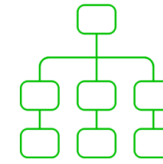
- Project Phases and Deliverables
- Project Work Breakdown Structure (WBS)
- Project Responsibility Assignment Matrix Formulation



**Project scope
and statement**



**Project
phases and
deliverables**



**Project work
breakdown
structure**

Project Activities, Phases and Deliverables

- A project task is “a defined unit of work”. A task can be considered as a detailed, lower-level **operation** carried out in a project to accomplish a high-level activity.
- A project **phase** can be defined as a “**stage**” in the life of a project.
- **Why do we define projects in phases?**
 - The future is **uncertain**.
 - We roughly plan the overall phases of a project, but **detail planning** is done one phase at a time.

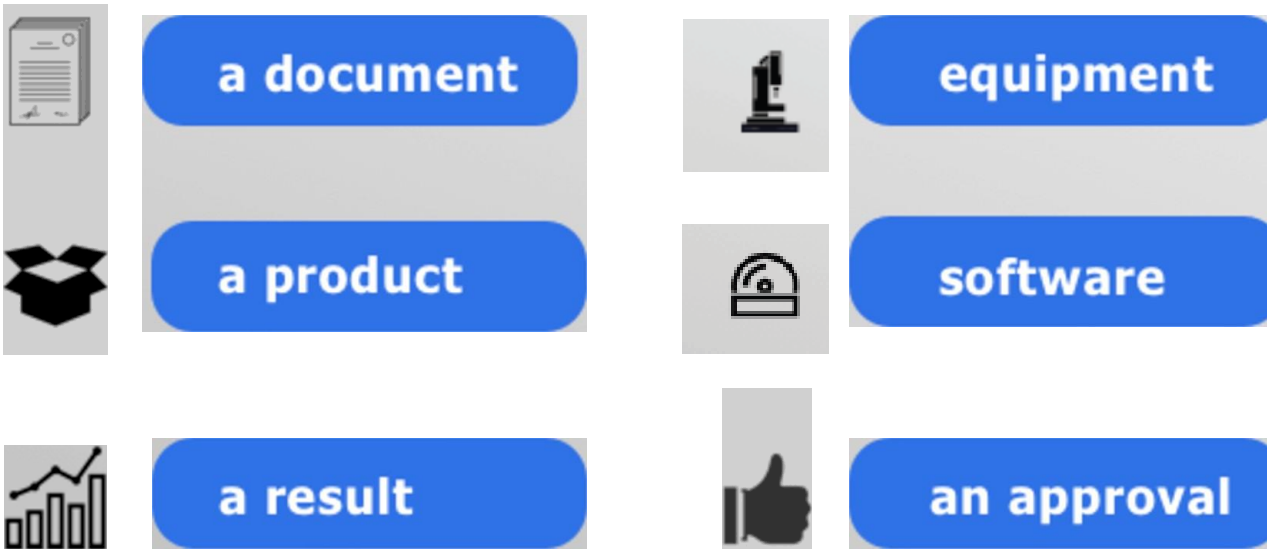
Project Phases and Deliverables

- Unnecessary expenditure of resources, time and effort can be avoided.

Deliverables are anticipated outputs in a project

- Every project phase may have a few high-level deliverables.
- A deliverable is a tangible thing-it is described using nouns and adjectives.

A PROJECT DELIVERABLE CAN BE...



More information and examples:

<https://www.tacticalprojectmanager.com/basics/what-are-project-deliverables/>

<https://www.workamajig.com/blog/project-deliverables-guide>

Work Breakdown Structure (WBS)

“... a structure that indicates decisions regarding how work is subdivided and assigned to different people (Steyn et al. 2016, pp. 81).”

- There are many variations of WBS. Use the methodology in these slides for the BSS 310 project.
- The UK approach uses verbs to describe the work breakdown, the US approach uses nouns. Either is acceptable in practice if used consistently, but stick to the UK approach for BSS 310.
- A rough WBS is advisable for each project phase to make sure you are not missing out or underestimating any critical dependencies.

Work Breakdown Structure (WBS) and its Significance

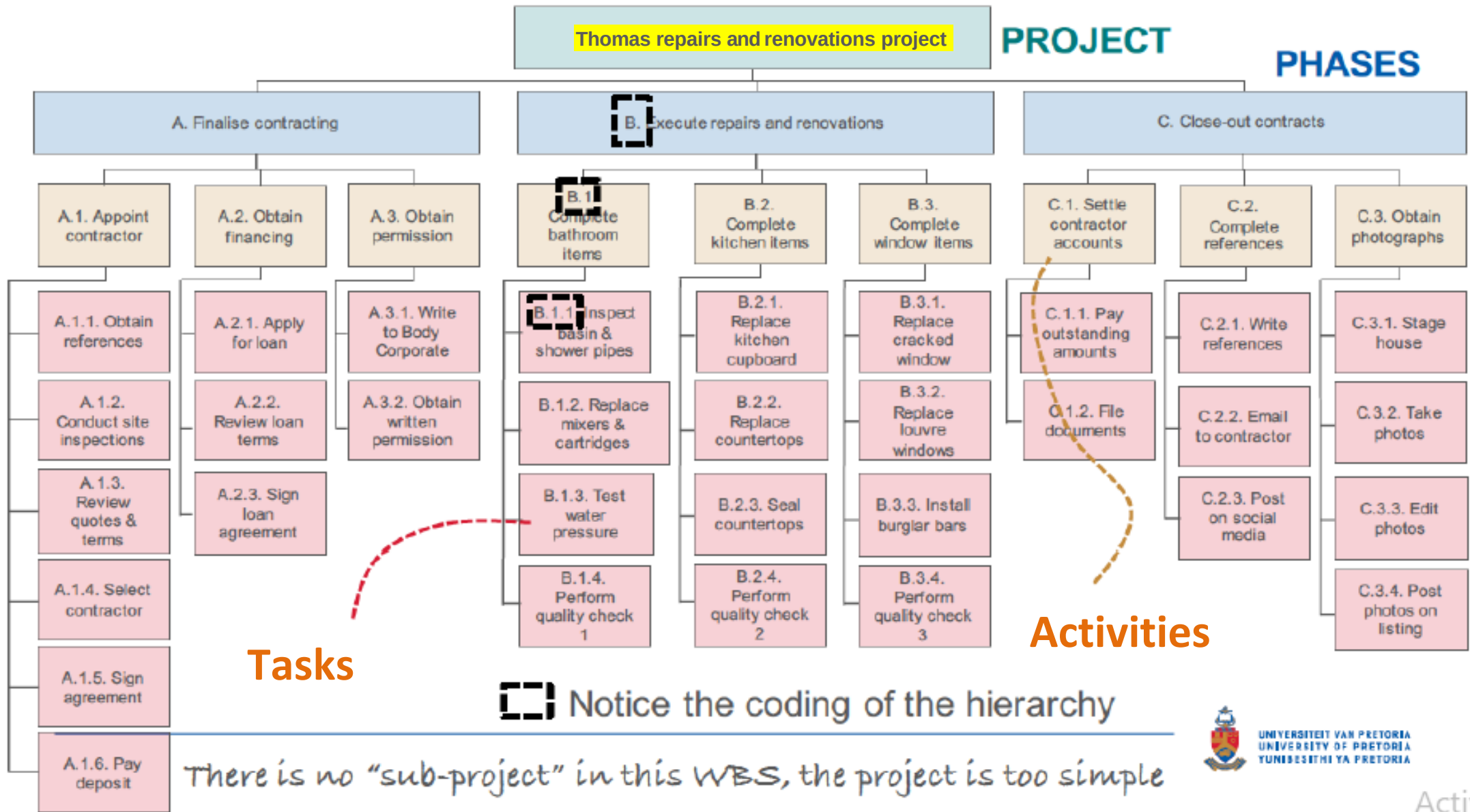
- Project is divided into parts in order to enable easier control and resource allocation
- Focus on deliverables
- Assignment of responsibilities
- Project budget and schedule can be based on WBS
- Monitoring and control of the project can be based on WBS
- Helps risk identification and management
- Project > Sub-projects (main areas/deliverables) » Project phases > Activities > Tasks

Simple Work Breakdown Structure (WBS) Template

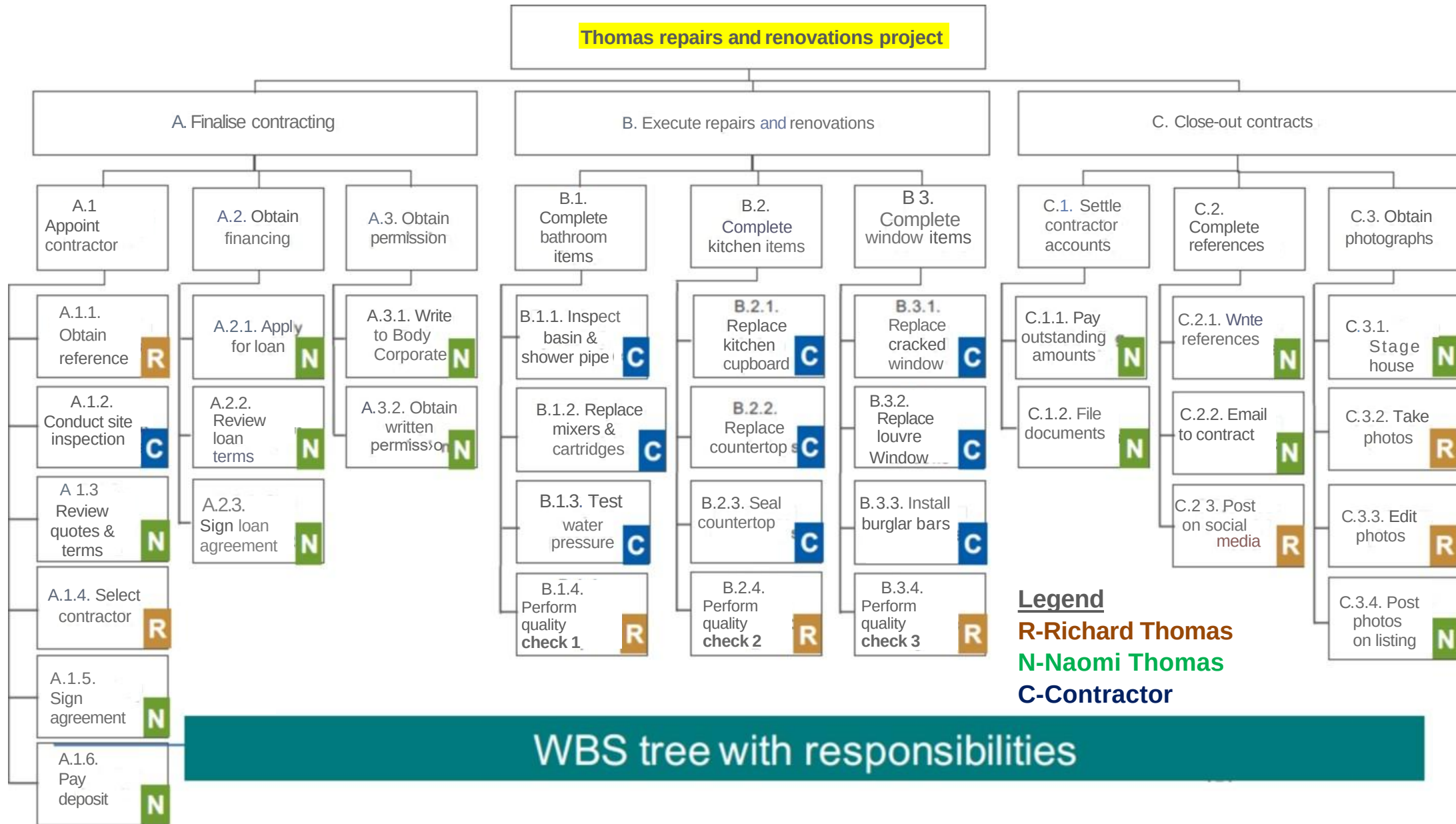
Project Title		Company Name	
Project Manager		Date	



Sample Work Breakdown Structural Tree



Work Breakdown Tree with Responsibilities



Project Responsibility Assignment Matrix

- **Responsible** - Those who do the work to achieve the task. There is typically one role with a participation type of responsibility, although others can be delegated to assist in the work required.
- **Accountable** - Sometimes also known as an "Approver" or final "Approving authority". This is the one ultimately accountable for the correct and thorough completion of the deliverables or tasks, and the one to whom the "Responsible" is accountable. In other words, an Accountable must sign off and (Approve) a work that a "Responsible" provides.
- **Consulted** - Those whose opinions are sought and with whom there is often a two-way technical communication.
- **Informed** - Those who are kept up-to-date on progress, often only on completion of the task or deliverable; and with whom there is often a one-way technical communication.

Project Responsibility Assignment Matrix

R = Responsible A = Accountable C = Consulted I = Informed

Activity	WBS Code	Project Team Members			Other Stakeholders		
		Mrs N Thomas (Owner)	Mr R Thomas (Owner)	Mr A Smit (Contractor)	Ms V Sheba (Body corporate)	Mr J Rossow (Neighbour)	
Obtain references	A.1.1.	I	A, R		C	C	Must be a name/person, not just organisation
Conduct site inspections	A.1.2.	I	A	R			There must always be an accountable <u>and</u> responsible person named for each activity!
Review quotes and terms	A.1.3.	R	A	C			
Select contractor	A.1.4.	C	A, R	I	I		

More information: <https://www.cio.com/article/2395825/project-management-how-to-design-a-successful-raci-project-plan.html>

Tips on Workload and Resources

- The 80/20 rule: most of the people spend their days in frenzy activities producing very little results because they are not concentrating on the right things
- Need for realistic planning and estimates of the hours or man-days of work required for activities as well as durations of the activities (especially in a multi project environment)
- Do not allocate 100% of your individual resources but reserve for other tasks and unexpected events
- Do not underestimate the work required for coordination, changes and rework
- Note that it is critical to have understanding of competence levels
- Individual productivity levels will certainly vary



More on Work Breakdown Structure:

<https://youtu.be/akO2Lf1fHmM>

**More on Project Responsibility
Assignment Matrix (RACI Chart):**

https://youtu.be/i9J_1N3-ibq

<https://miro.com/templates/work-breakdown-structure-basic/>

<https://www.smartsheet.com/free-work-breakdown-structure-templates?srsId=AfmBOoohA37itX1aU-KEDgNEcmmWjfraZQdPzRgrOMLGMIC11ue9MiBc>

Learning Outcomes Evaluation

- Did you learn something new from this lecture in respect of project management?
- Can you at this point comfortably explain the concepts of project, sub-project, phases, activities, tasks and deliverables using your own words?
- Can you at this point independently create a Work Breakdown Structure (WBS) for a project? Including carrying out an analysis and evaluation?
- Can you create a Project Responsibility Assignment Matrix for the semester project?

The End

Next lecture

Unit 9: Network diagram and related metrics



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